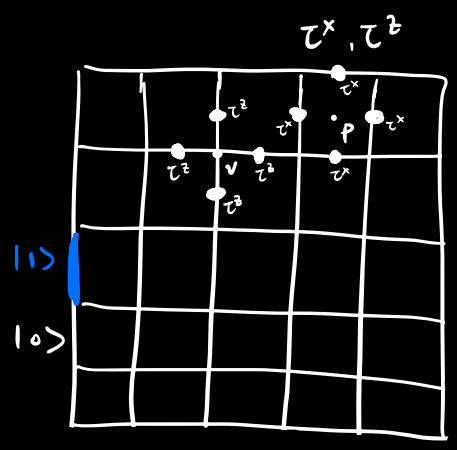
# Topological lattice model from Gauging

local symmetry
(constrain)

## Toric Code and $\mathbb{Z}_2$ gauge theory




$$H = -\sum_v A_v - \sum_p B_p$$

- $[A_v, A_{v'}] = [B_p, B_{p'}] = 0$

  $[A_v, B_p] = 0$

$\Longrightarrow$ 局部地找 $A_v$, $B_p$ term 的本征态，即为 H 的本征态；局部地找 ground state，即为 global ground state.

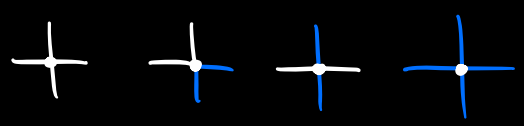
因此，只需局部地分析即可.

- pauli 算符平方 $=1$ $\Longrightarrow$ 本征值 $\pm 1$. 因此基态是本征值为 $+1$ 的态.

能量最低

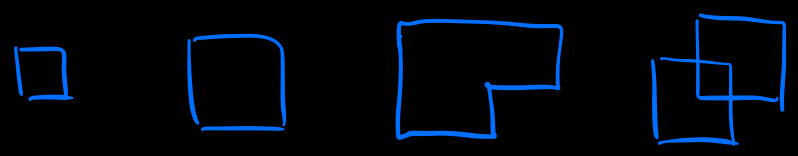
### 考虑 $A_v$ 的要求：

$$\begin{cases} \tau^z |0\rangle = 1\,|0\rangle & \text{not occupied by a string} \\ \tau^z |1\rangle = -1\,|1\rangle & \text{occupied by a string} \end{cases}$$

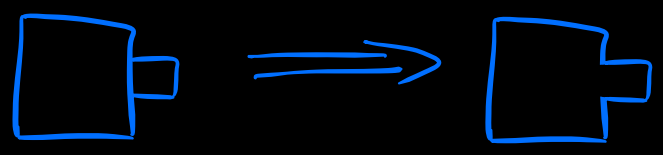
$\Longrightarrow$ we need  and their rotationally symmetric configurations.

一个 vertex 上有偶数个 string $\Longrightarrow$ 一进一出 $\Longrightarrow$ 形成 loops

i.e. $A_v \Longrightarrow$ loops

Rmk  different shapes

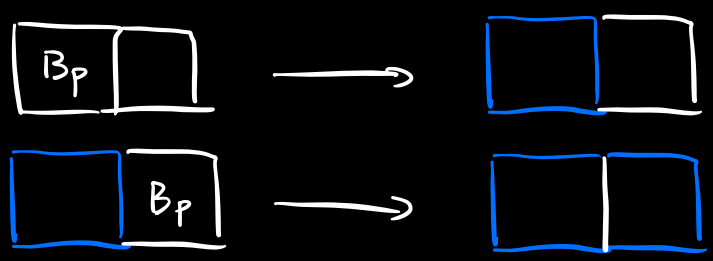
if two loops share boundaries:



($\mathbb{Z}_2$, $-+-=-$)

## 考虑 $B_p$ 的要求:

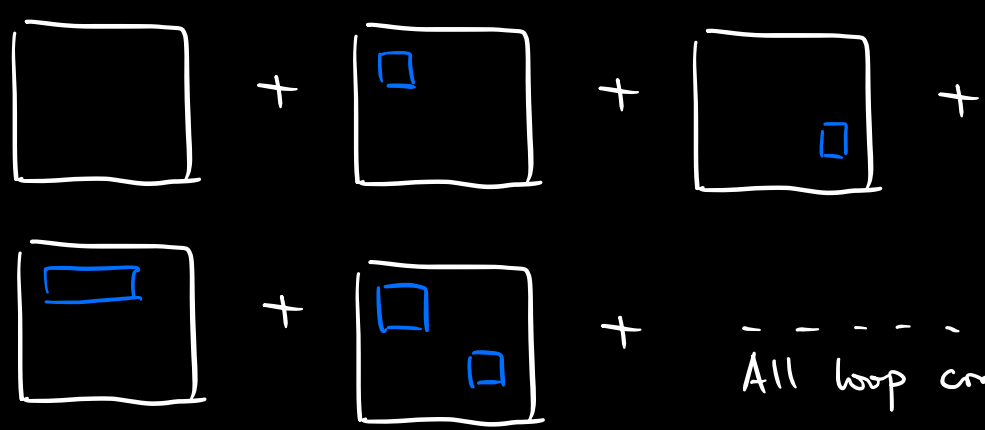
在一个无弦的 lattice 中, $B_p$ 会 create 一个 loop:




(flip twice)

$B_p|\psi\rangle = |\psi\rangle$ (+1 eigenvalue)

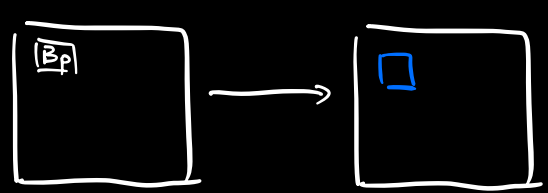
↳ ground state wavefunction

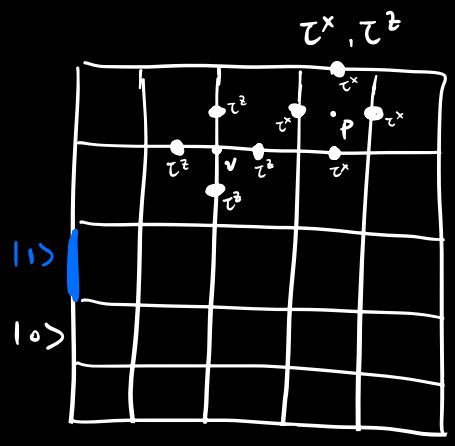
$|\psi\rangle =$ 

All loop configurations

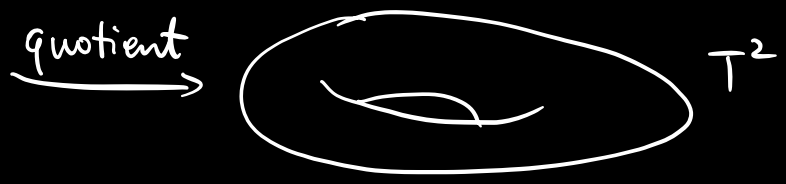
Rmk 在所有 configurations 中, $B_p$ 可以在其中将一个相映射到另一个相中.

e.g.

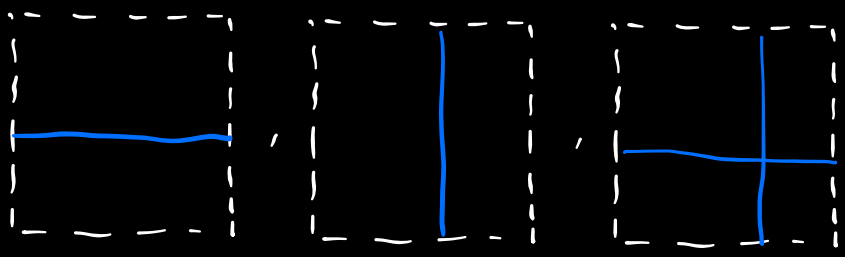




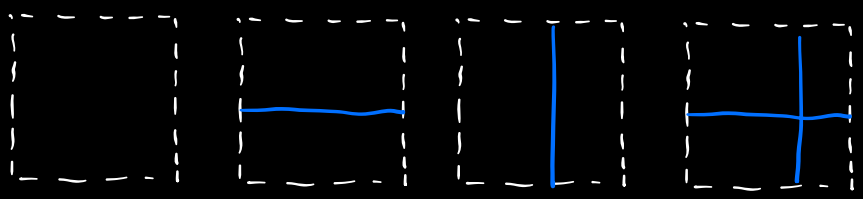

若将此二维系统的四个边界 quotient 掉使其成为一个 Torus，情形就不同了！




等价地，可以说采用 periodic boundary condition (PBC)

此时，  也是 loops.

这种 loop 不能 created by $B_p$.



从这四种情形出发，我们可以用 $B_p$ 在其中 create 那些可缩的 loops；其态的 Hilbert space 分成四个子空间，即：

在 上，有 4-fold ground state degeneracy.

* 在 Sphere $S^2$ 上，仅有 1-fold，因为球面上任一 loop 均可缩 (Non-degenerate)

<u>Rmk</u> 2 constrains: $\underbrace{\prod_v A_v = \mathbb{1}, \quad \prod_p B_p = \mathbb{1}}$

Dimension of total Hilbert space $4^{N_v}$

2 Hamiltonian terms per unit cell $\Rightarrow 4^{N_v}/(4^{N_v}/4) = 4$

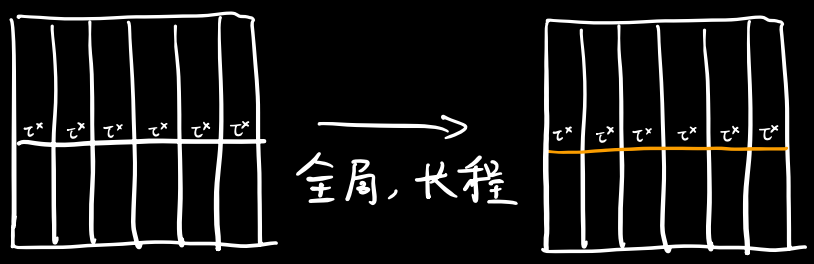
<u>Rmk</u> 4-fold 简并对 $\forall$ local 微扰是 stable 的.

* 为了说明这点，需要考查如何实现不同基态间的映射，(e.g. )

发现这需要 global，long-range 的操作，局部微扰办不到.

Logical operator : 实现不同 ground states 间的 maps.

e.g.




$$\begin{cases} H|\psi_1\rangle = E_{min}|\psi_1\rangle \\ H|\psi_2\rangle = E_{min}|\psi_2\rangle \end{cases}$$

$$\langle\psi_1|\prod_L \tau^x|\psi_2\rangle = 1$$

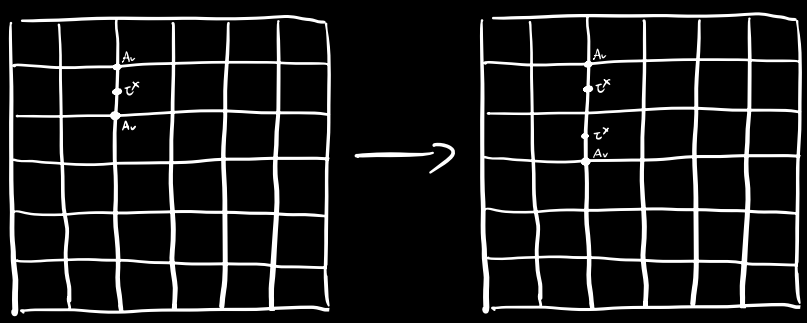
$$H + \varepsilon\sum_i \tau^x$$

$$\langle\psi_1|\tau^x|\psi_2\rangle = 0$$

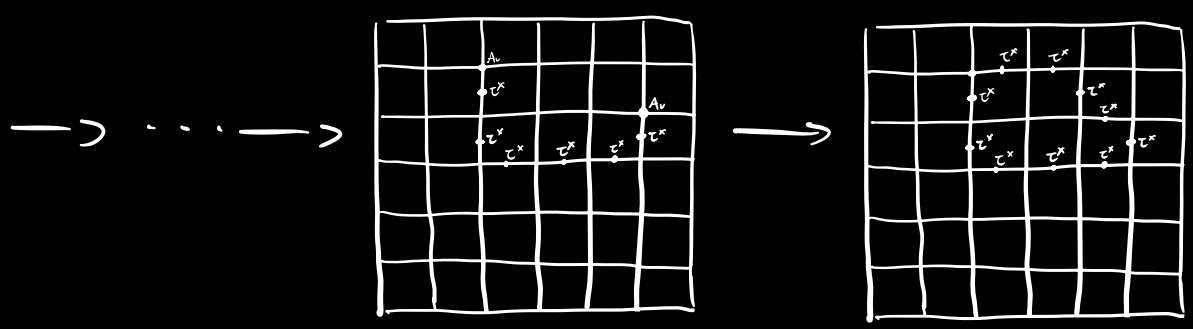
$$\langle\psi_1|\varepsilon^L\prod_L \tau^x|\psi_2\rangle \propto \varepsilon^L$$

## Excitation:

$\tau^x$ :




$\tau^x$ 创造了两个顶点的激发，随后用 $\tau^x$ 作用可以将这两个激发分开




若走一个loop，则回到基态.

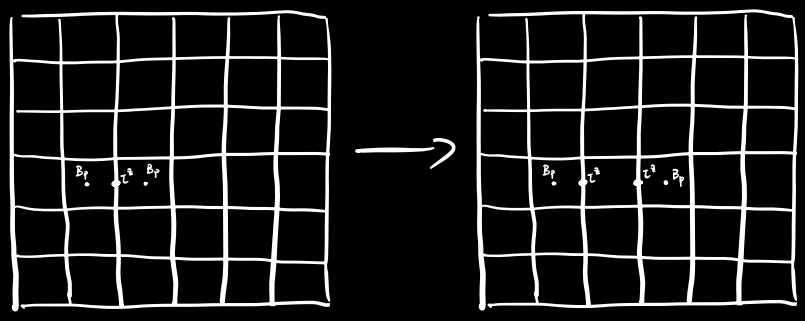
Violation of $A_v$ $\Longrightarrow$ called a charge excitation

$$* \begin{cases} B_p\tau^x|\psi\rangle = \tau^x B_p|\psi\rangle = \tau^x|\psi\rangle \\ A_v\tau^x|\psi\rangle = -\tau^x A_v|\psi\rangle = -\tau^x|\psi\rangle \end{cases}, \quad |\psi\rangle: \text{ground state}$$

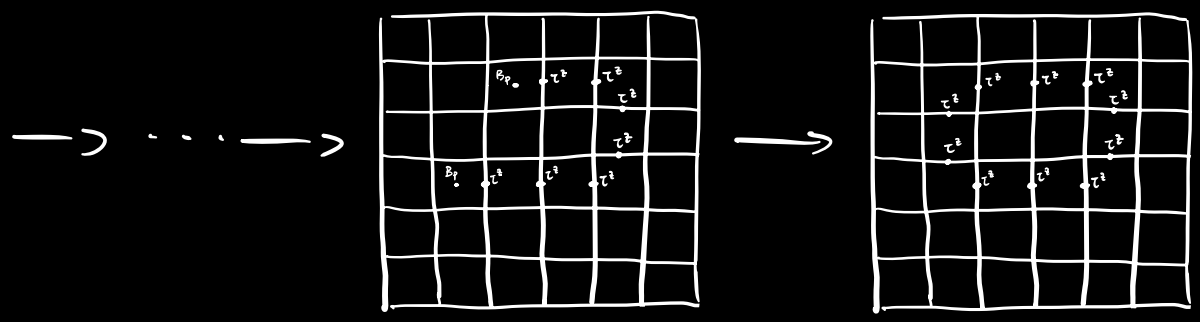
( $A_v$ is excited)

$\tau^z$ :

$$\begin{cases} A_v\tau^z|\psi\rangle = \tau^z A_v|\psi\rangle = \tau^z|\psi\rangle \\ B_p\tau^z|\psi\rangle = -\tau^z B_p|\psi\rangle = -\tau^z|\psi\rangle \end{cases}, \quad |\psi\rangle: \text{ground state}$$




$\tau^z$ 创造了两个顶点的激发，随后用 $\tau^z$ 作用可以将这两个激发分开




若走一个loop，则回到基态

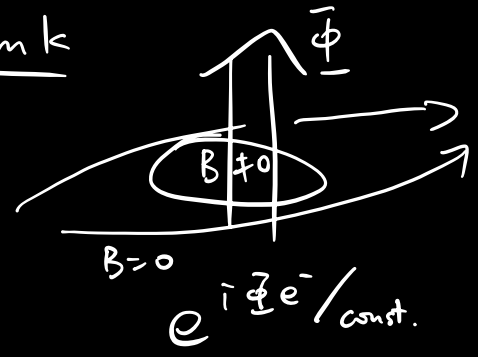
Violation of $B_p$ $\Longrightarrow$ called a flux excitation

* Charge, flux excitation are the two basis excitation of toric code.

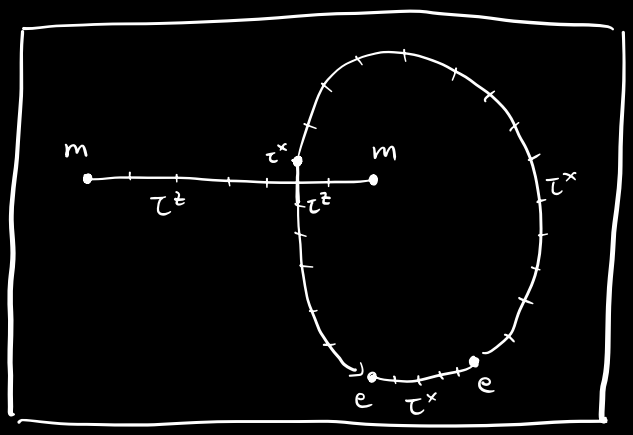
| notation |

e m

Rmk




猜测




The Aharonov-Bohm Effect

转一圈差一个相位 (Berry phase)

果然，由于 $\tau^x$ 和 $\tau^z$ 反对易，先绕圈后造 m 和先造 m 后绕圈差一个负号。

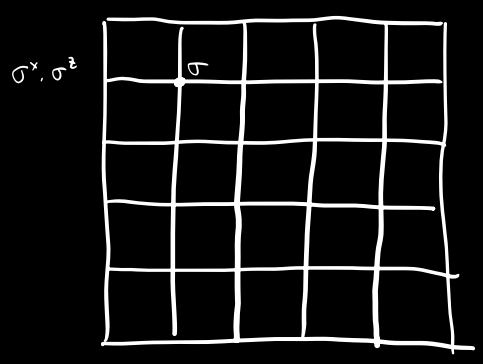
因此这里也有 A-B effect.

$e^{i\pi}$

## Rmk

$\Longrightarrow$ e, m are bosons.

## Transverse field Ising model




$$H = -J \sum_{\langle i,j \rangle} \sigma_i^x \sigma_j^x - \sum_i \sigma_i^z$$

$$U = \prod \sigma_i^z \qquad HU = UH$$

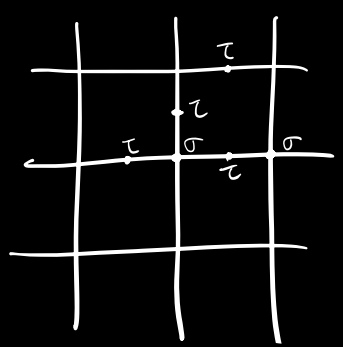
$J << 1$ symmetric phase

$J >> 1$ symmetric breaking phase

Couple TFI model to $\mathbb{Z}_2$ Gauge field

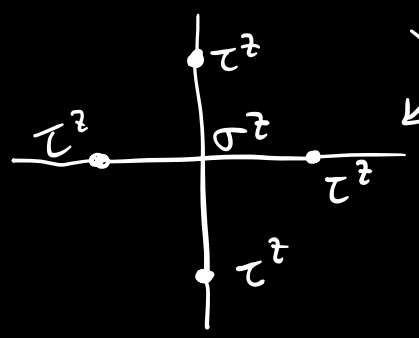
|

local



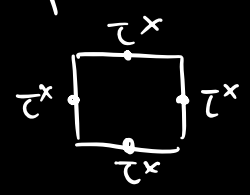

we want to couple them together to get local symmetry.

local symmetry operator $\quad (\mathbb{Z}_2 : \ \cdot^2 = \mathbb{1})$




是这5个家伙的 tensor product

zero-flux condition




$$\rightsquigarrow \quad H_g = -J \sum_{\langle i,j \rangle} \sigma_i^x \tau_{\langle i,j \rangle}^x \sigma_j^x - \sum_i \sigma_i^z - \sum_p B_p$$

↳ 消除简并

$J=0$: $H_g = -\sum_i \sigma_i^z - \sum_p B_p$

$$= -\sum_v A_v - \sum_p B_p$$

local symmetry 的存在，使所有东西心结是它的 eigenstate，因此有

Local Sym $|\sigma^z\rangle = |\sigma^z\rangle$ (eigenfunction)

$= |\tau^z\rangle$ (since $\sigma^{x^2} = \mathbb{1}$)

$\rightsquigarrow$ 正是 $A_v$

Sym. phase of TFI model with global $\mathbb{Z}_2$ - sym

$\Downarrow$ 与 $\mathbb{Z}_2$ - gauge field 耦合，使得 model becomes locally symmetric.

Toric code with topological order

## Summary

1. Dynamics for matter field is the same at low energy.
2. Gauge charge comes from symmetry charge. $\quad e \times e = 1 \longrightarrow$ sym.
   ↳ bosonic $\longrightarrow$ statistic of sym. charge
3. AB phase $\longrightarrow$ determined by sym. group.